Activity 3

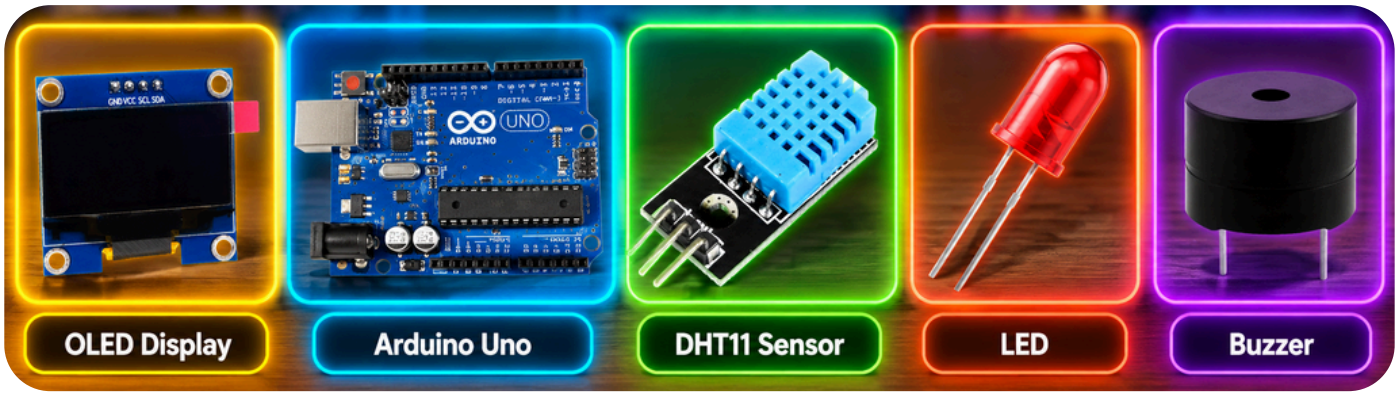
Activity Name: Humidity Alert System



The Problem: Riya noticed that the classroom sometimes felt too humid and uncomfortable, especially during rainy days. She wondered how she could know when the humidity level becomes too high. To solve this problem, she decided to build a humidity alert system that can measure and display humidity. The system will also turn on an LED and buzzer when the humidity reaches a high level.

What Will We Build: We will build a humidity alert system that measures the humidity in the air. It will show the humidity on an OLED display and turn on the LED and buzzer when humidity is too high.

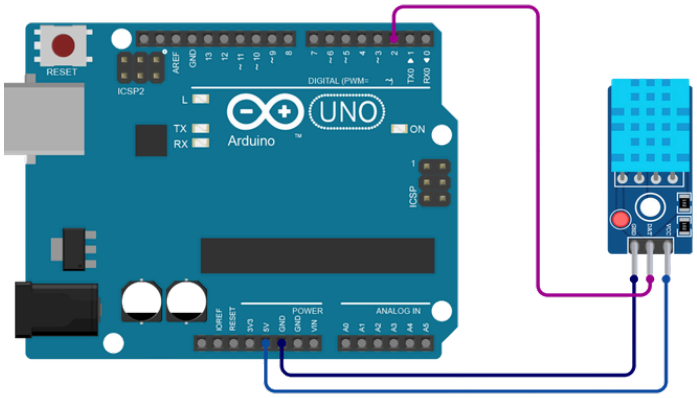
Components Required



Circuit Designing

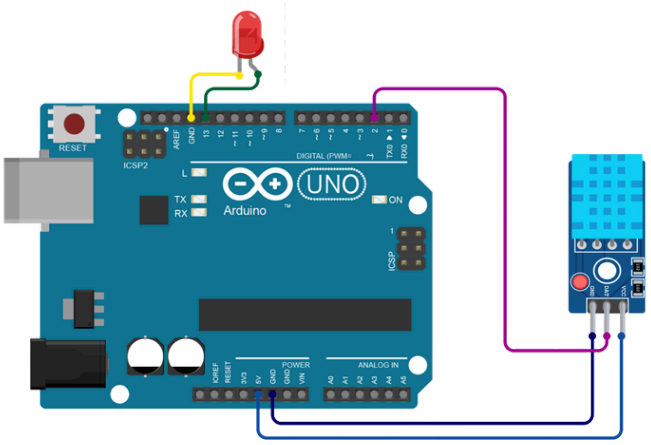
Step 1: Connect the DHT11 Sensor to the Arduino Uno.

Sensor Pins	Arduino Pins
Vcc Pin	5V Pin
GND Pin	GND Pin
Data Pin	Digital Pin 2



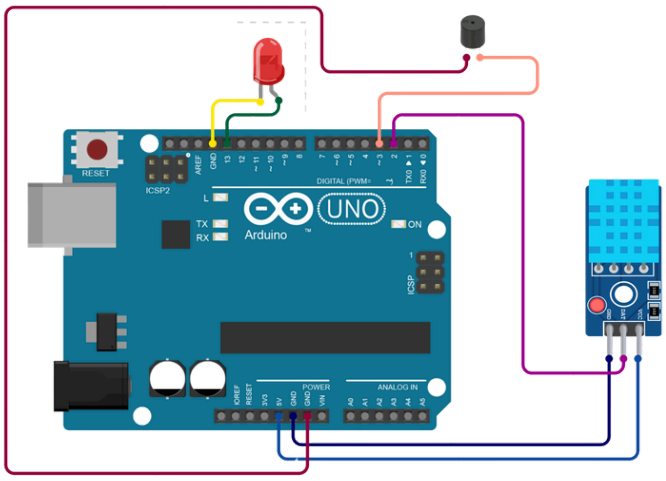
Step 2: Connect LED to the Arduino Uno.

LED Pins	Arduino Pins
Positive Pin	Digital Pin 13
Negative Pin	GND Pin



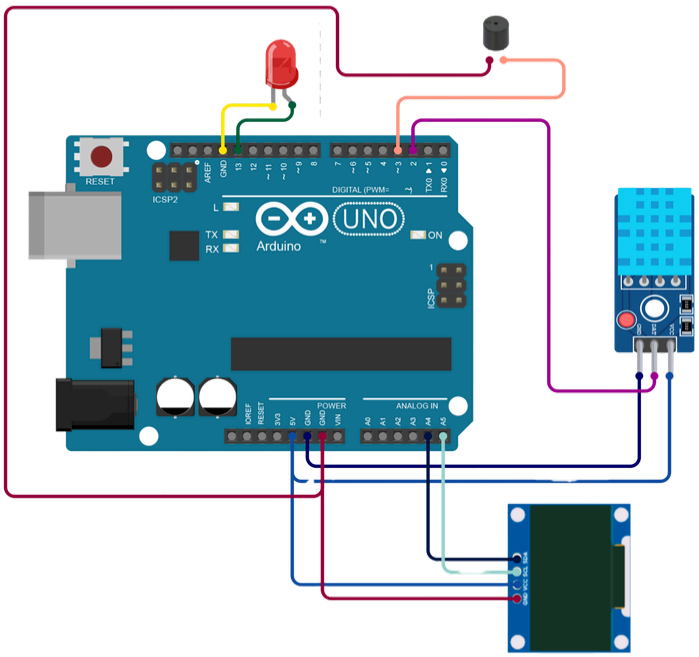
Step 3: Connect buzzer to the Arduino Uno.

Buzzer Pins	Arduino Pins
Positive Pin	Digital Pin 3
Negative Pin	GND Pin



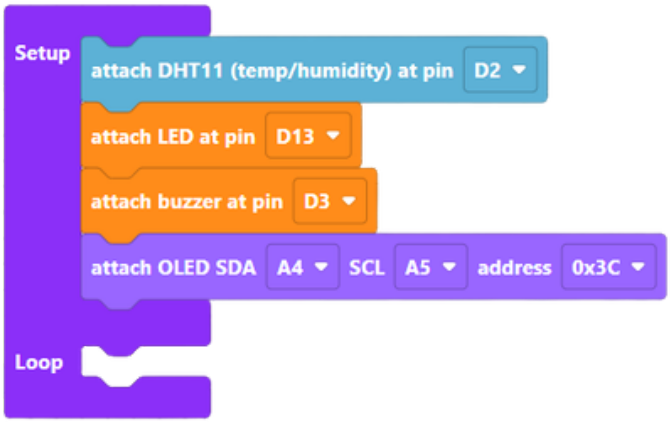
Step 4: Connect the OLED display to the Arduino Uno.

Display Pins	Arduino Pins
Vcc Pin	5V Pin
GND Pin	GND Pin
SDA Pin	Analog A4 Pin
SCL Pin	Analog A5 Pin

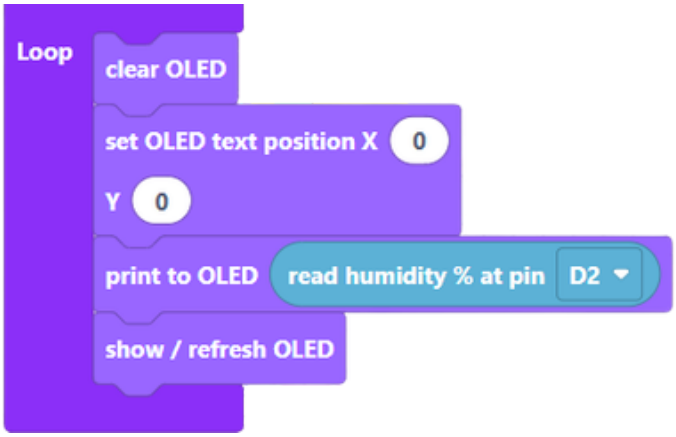


Project Coding

Step 1: Make code to initialize the pins



Step 2: Make code to display the humidity value on the OLED display.



Step 3: Make code to use the conditions to check the humidity level.



Step 4: Use the delay in milliseconds for light on and off



Learning Outcomes

- The students will be able to identify the DHT11 sensor and understand that it measures humidity.
- The students will be able to connect the DHT11, LED, buzzer, and OLED to the Arduino Uno.
- The students will be able to display the humidity value on an OLED display using blocks.
- The students will be able to use an if-else condition to turn the LED and buzzer ON when humidity is above 60%.



BRAIN TEASERS

Section A – Multiple Choice Questions

1. Which sensor is used to measure humidity?

A. LED
B. DHT11 Sensor
C. Buzzer
D. OLED Display

2. What happens when the humidity is above 60%?

A. The LED and buzzer turn ON
B. The OLED turns OFF
C. The Arduino turns OFF
D. Nothing happens

3. Which device shows the humidity value?

A. Buzzer
B. LED
C. OLED Display
D. Arduino cable

Section B – True or False

1. The DHT11 sensor is used to measure humidity.....

2. The buzzer is connected to Digital Pin 13.....

3. The OLED display can show the humidity value.....